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The influence of economic development level, household wealth and maternal education on child health in the developing world

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Abstract

This study estimates the relative importance to child health (indicated by weight and height for age) of economic development level [gross domestic product (GDP) converted to international dollars using purchasing power parity (PPP) rates: GDP-PPP], household wealth and maternal education and examines the modifying influence of national contexts on these estimates. It uses information collected from mothers aged 15–49-years participating in Demographic Health Surveys (DHS) conducted in 42 developing countries. In multilevel regression models, the three study variables exhibited strong independent associations with child health: GDP-PPP accounted for the largest amount of unique variation, followed by maternal education and household wealth. There was also substantial overlap (shared variance) between maternal education and the other two study variables. The regressions of child health on household wealth and maternal education exhibited substantial cross-national variation in both strength and form of association. Although higher education levels were associated with disproportionately greater returns to child health, the pattern for household wealth was erratic: in many countries there were diminishing returns to child health at higher levels of household wealth. We conclude that there are inextricable links among different strategies for improving child health and that policy planners, associating benefits with these strategies, must take into account the strong moderating impact of national context.
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Introduction

A number of economic and social strategies have been proposed for improving child health in the developing world (Dreze & Sen, 1989; Pritchett & Summers, 1996). These include economic development, reduction of poverty and the provision of educational, health and social welfare programs. Many of these strategies are interdependent and

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their effects subject to national forces operating within host countries. It is important to disentangle the unique impact of each strategy and estimate to what extent this impact is modified by country specific circumstances. This would help policy makers understand both the leverage available and the context required for these strategies to achieve their child health objectives.

This report examines the influence of economic development level (and by extension, the distribution of household wealth) and women's (maternal) education on child health. Economic development uses market forces to improve material circumstances in the population (e.g., quality of food, water and shelter); to provide adults with work opportunities and increased income; and to create capital for investment in the public and private sectors. Cross-national comparison studies and time-series analyses have confirmed strong associations between child mortality and malnutrition in developing countries, and economic indicators such as gross national product (GNP) and gross domestic product (GDP) per capita (Alderman, Appleton, Haddad, Song, & Yisehac, 2001; Filmer & Pritchett, 1999a; Pritchett & Summers, 1996; Shen & Williamson, 2001; Smith & Haddad, 2001; Wang, 2003).

Household wealth offers leverage for improving child health within countries in much the same way that economic development level can improve child health nationally. It provides an opportunity to improve the material circumstances of the family and to purchase goods and services that are health enhancing. It is well established at the low end of the wealth spectrum that poverty is an important determinant of mortality and poor health in all countries (WHO, 1999). Furthermore, the evidence suggests that there are likely to be positive monotonic relationships between household wealth and child health in most developing countries (Gwatkin, 2001).

Women's education offers leverage for improving child health through the provision of more effective parental care in the home and enhanced use of treatment and prevention services from the health care system (Caldwell, 1979, 1994). It may also contribute to delayed child bearing, longer birth intervals and fewer children as well as improved opportunities to pursue work outside the home and to generate additional household wealth. Empirical studies show repeatedly that women's education is associated with longer life expectancies, lower death rates and improved child health and nutrition (Barrera, 1990; Buchmann, 1996; Hadden & London, 1996; Schultz, 2002). However, some researchers argue that these associations are much weaker when appropriate statistical controls are introduced (Basu, 1994; Hobcraft, 1993). Desai and Alva (1998), for example, in separate analyses of 22 developing countries, showed that controlling for paternal education, residency and access to piped water and toilet facilities attenuated the association between maternal education and child height for age by about 67%. The authors concluded that maternal education acted as "…a proxy for socioeconomic status of the family and geographic area of residence" (p. 71).

An accurate estimation of the impact on child health of economic development level, household wealth and woman's education requires: (1) quantifying the unique and shared influence attributable to each, and (2) accounting for the moderating effects of national contexts—both, fundamental challenges. At a theoretical level, the challenge of attribution—decomposing the influence of interdependent strategies on health—is substantial. Governments invariably use resources generated by economic development to improve literacy and educational levels. Improved literacy and educational levels can enhance parental care-giving which directly impacts child health. In this instance, education acts as a mediating variable or pathway through which economic development has a positive influence on parenting and child health. Of course, education is not the only factor that influences parenting. When economic development increases household wealth and access to resources and services, and this leads to improved maternal health and parental care-giving, then the mediating variable or pathway through which economic development level has a positive influence on parenting and child health is wealth creation, not education.

The effects of increasing household wealth and educational levels are not only 'downstream' towards enhanced parenting and child health but also 'upstream' towards economic development. Education is the primary means for creating a skilled work force; that, in turn, provides the human capital to sustain economic development through innovation and increased productivity. Household wealth can contribute to this process as well if it enhances the health of workers. Both household wealth and women's education sit in the middle of a causal path: on one hand, they mediate some of the

child health benefits associated with economic development; on the other, they provide a basis for sustaining economic development processes (Ranis, Stewart, & Ramirez, 2000). In the absence of experimental research, efforts to disentangle the impact of these strategies must acknowledge their interdependence, both distally and proximally, and rely on statistical approaches for insight into their relative importance.

The second fundamental challenge is to account for the moderating effects of national contexts. There is no doubt that the influence of economic development level, household wealth and women's education on child health will depend on the national contexts in which they operate. For example, Alderman et al. (2001) demonstrated that income growth is associated with lower levels of malnutrition in 12 developing countries, but also noted that the impact of increased household income per capita on child height for age was weaker for the two low-income African countries. Several authors, working in a limited number of countries, have noted anomalous findings that highlight cross-national variation. For example, life expectancies are much higher in Sri Lanka and the Indian state of Kerala than would be expected based on national income; and the reverse is true in Brazil—much lower life expectancy than would be predicted by income growth (Anand & Ravallion, 1993; Dreze & Sen, 1989; Sen, 2001).

Based on evidence of diminishing returns to life expectancy associated with increased income (Subramanian, Belli, & Kawachi, 2002), it is reasonable to expect that some developing countries will exhibit a concave relationship between household wealth and child health. In other countries, however, there may be no relationship: this was found in three of nine developing countries when Wagstaff (2000) examined associations between child mortality and five levels of household consumption. Finally, Desai and Alva (1998), in their separate analysis of 22 developing countries, reported large cross-national differences in the magnitude of associations between maternal education and child height for age. National context is thus an important consideration when estimating the impact of strategies to improve child health.

Separating out the true causal impacts on child health of economic development level, household wealth and women's education requires information that may never be available to researchers. There are, however, methods in regression analysis for estimating explained variance associated with predictor variables and then using nested models to divide this explained variance into unique and shared portions. Distinguishing between unique and shared variance attributable to economic development level, household wealth and women's education will not solve the attribution problem but will provide some insight into the relative contributions to health provided by these strategies. Furthermore, ordinary least square regression models have been extended to permit the correct analysis of correlated data structures—where individuals may be nested in meaningful groupings such as households, communities and countries. These multilevel regression models disaggregate residual variation between-and within groupings of a hierarchy (Kreft & deLeeuw, 1998; Snijders & Bosker, 1999). In the present study, these models allow direct estimation of the impact of national contexts on within-country associations between child health and postulated influences such as household wealth and women's education.

We will address five questions. One, to what extent is variability in child health associated with between-versus within-country differences? Two, what is the strength of association between child health, economic development level, household wealth and maternal education? Three, is there evidence of between-country differences in the strength and form of association between child health and household wealth and between child health and maternal education? Four, how much unique and shared variability in child health is associated with economic development level, household wealth and maternal education? Five, what percent of cross-national variability in child health is associated uniquely with economic development level?

Method

The data come from single, cross-sectional Demographic Health Surveys (DHS) conducted in 42 developing countries between 1994 and 2003. The DHS program began in 1984 and has conducted a total of 104 surveys (some repeated at 3–6 year intervals) in 62 developing countries. It strives to collect information in participating countries from nationally representative probability samples of individuals using area-based sampling frames to select intact clusters and households (Macro International Incorporated, 1996). A stratified sample of

areal units, usually enumeration areas, is divided into equal size segments of 500, and selected with probability proportional in size to the number of segments they contain. In areas with more than one segment, one is selected at random to yield a single-stage, equal probability sample of segments equivalent in size. In selected segments, all households are listed and a fixed proportion, selected by systematic sampling. These sampled areas are called 'clusters'. In rural areas, approximately 30–40 households are sampled from each cluster; in urban areas, approximately 20–25 households are sampled. The samples are self-weighting in most countries: where sampling weights have been derived, they have had little impact on point estimates. Enlistment into the DHS studies is very high: participation rarely goes below 90% of sampled households. Extremely remote or dangerous areas may be excluded systematically as well as families not living in some form of dwelling. Standard data collection methods are used to obtain health, socio-economic and fertility-related information from women 15–49 years of age living in household dwellings.

Concepts and measures

Child height and weight for age. Child height and weight are based on physical measurements taken by interviewers using a standard protocol. These measures are available for each child up to 59 months of age in 29 countries and up to 35 months of age in the remaining 13 countries; the anthropometric measures were converted to standard scores (z-scores) using the US National Child Health Survey anthropometric data as the international reference.

Economic development level, household wealth and maternal education. At the national level, economic development level is measured by the log of GDP per capita (GDP) based on purchasing power parity (PPP). GDP-PPP is gross domestic product converted to international dollars using PPP rates (World Bank, 2004). Information to measure parent and household variables is based on maternal responses to a structured interview. Household wealth was measured by the asset index derived from consumer durables (e.g., radio and TV) and dwelling conditions (e.g., drinking water source) associated with each household (Filmer & Pritchett, 1999b, 2001). This index is formally defined as the largest principal component of a set of binary asset variables; it is a linear combination of these variables weighted by component-score weights that maximally account for their co-variation. It was developed to measure household wealth within but not between countries, and available evidence suggests that it performs as well as consumption expenditures (Bollen, Glanville, & Stecklov, 2002; Houweling, Kunst, & Mackenbach, 2003). In households comprised of multiple mothers, all were assigned the same asset index. Maternal education was based on total years of schooling and was measured individually for each mother.

Control variables. The following covariates served as control variables in all analyses: child sex (0, female; versus 1, male), child age in months (0–59), child age in months squared; maternal age in years and maternal age in years squared. No attempt was made to control for other potential covariates such as marital status, family structure, household composition and residency. The precise causal role of these factors on child health is unclear, and it is arguable that most of these factors are subject to influences arising from economic development level, household wealth or women's education.

Analysis

The study uses multilevel modeling and the statistical software MLwiN (Rasbash, Steele, Browne, & Prosser, 2004) to conduct a four-level regression analysis of the DHS data with simultaneous consideration of i children, nested in j households, nested in k clusters, nested in f countries. Separate analysis indicated that the variability arising from having two or more mothers nested in the same household was too small to be modeled as a separate level. As a result, children in households born to different mothers were modeled statistically as part of the same household: maternal age and education were treated as child level variables.

In multilevel regression, data structure—the correlation of responses within sampled groupings—is modeled explicitly. This is achieved by disaggregating residual errors associated with each group or level in the hierarchy and expressing these errors as patterns of variation. Eq. (1) depicts a multilevel 'null' model with no explanatory variables. There are four levels: i, j, k and f, corresponding to children, households, clusters and countries. In this model, y_{ijkf} represents the health of child i, from household j, in cluster k, in country f; $\beta_0 x_0$ represents the fixed effect intercept,

i.e., the overall

$$y_{ijkf} = \beta_0 x_0 + (v_f + u_{kf} + w_{jkf} + \varepsilon_{ijkf}) \quad (1)$$

average level of child health in the total population. The terms inside the brackets are residual errors, for example, v_f is the difference in child health between the average response of country f and the intercept, $\beta_0 x_0$. In Eq. (1), v_f, u_{kf}, w_{jkf}, ε_{ijkf} are random quantities; they are assumed to be uncorrelated across levels; and distributed normally with 0.0 means and variances, $\sigma^2{}_v$, $\sigma^2{}_u$, $\sigma^2{}_w$ and $\sigma^2{}_e$, respectively. The important point is that residual errors expressed as random effects variances characterize the distribution of response associated with each sampling level around the intercept, $\beta_0 x_0$: they provide the means to estimate intra-class correlation coefficients (ICC)—parameters that quantify the extent to which individual responses within a group are correlated. The ICC for country is given by $v_f/(v_f + u_{kf} + w_{jkf} + \varepsilon_{ijkf})$ and is expressed as a percent from 0.0 to 100.0.

Eq. (2) depicts the full model used in this study. The response variable y_{ijkf}

$$\begin{aligned} y_{ijkf} = {} & \beta_0 x_{ijkf} + \beta_1 x_1 \,\text{GDP-PPP}_f + \beta_2 x_2 \,\text{wealth}_{jkf} \\ & + \beta_3 x_3 \,\text{wealth-squared}_{jkf} + \beta_4 x_4 \,\text{education}_{ijkf} \\ & + \beta_5 x_5 \,\text{education-squared}_{ijkf} + \beta_6 x_6 \,\text{chsex}_{ijkf} \\ & + \beta_7 x_7 \,\text{chage}_{ijkf} + \beta_8 x_8 \,\text{chage-squared}_{ijkf} \\ & + \beta_9 x_9 \,\text{matage}_{ijkf} + \beta_{10} x_{10} \,\text{matage-squared}_{ijkf} \\ & + [\sigma^2_{v0} + (\sigma^2_{v2} + \sigma^2_{v3} + \sigma^2_{v4} + \sigma^2_{v5}) + \sigma^2_{u0} \\ & + \sigma^2_{w0} + \sigma^2_{e0}] \end{aligned} \quad (2)$$

is from Eq. (1). The top five lines present the fixed effects estimates. $\beta_0 x_{ijkf}$ represents the fixed effect intercept (i.e., the overall average level of child health). The β coefficients quantify the strength of association between dependent and independent variables measured throughout the hierarchy: they are interpretable as regression coefficients, specifying the change in the response variable associated with a unit change in the predictor variable, controlling for other variables in the model. GDP-PPP, wealth and education are assessed at the country, household and child level, respectively. Quadratic terms for wealth and education are included to test for non-linearity in their associations with child health. The control variables, child sex, child age, child age squared, maternal age and maternal age squared, are assessed at the child level.

The bottom two lines in Eq. (2) presents the random effects estimates. The random effects intercepts—$\sigma^2{}_{v0}$, $\sigma^2{}_{u0}$, $\sigma^2{}_{w0}$, $\sigma^2{}_{e0}$—represent the variability in child health at each sampling level (country, cluster, household and child). The random effects regression coefficients—$\sigma^2{}_{v2}$, $\sigma^2{}_{v3}$, $\sigma^2{}_{v4}$, $\sigma^2{}_{v5}$—represent between-country variability in the regression lines (slopes) between child health and wealth (β_2), wealth squared (β_3) education (β_4), and education squared (β_5). Each of the random effects co-variances—associations between random effects intercepts at the country level and random effects regression coefficients at the country level—were small in magnitude, statistically non-significant and excluded from the model.

Question 1 will be addressed by examining the random effects variances associated with the multilevel null model specified in Eq. (1). This question sees cross-national variation at level four expressed as a percent of variation across all levels. Questions 2–5 will be addressed by the full model specified in Eq. (2). Question 2 examines the fixed effects regression coefficients associated with GDP-PPP, household wealth and maternal education. This question draws attention to the strength of association between child health and the predictor variables expressed in their specific measurement units. Question 3 examines the random effects regression coefficients for household wealth and maternal education and focuses attention on level 4. Question 4 will be addressed by examining differences between the full model specified in Eq. (2) and nested models subsumed by the full model. In this question, the impact of the predictor variables is estimated in the common metric of unique and shared variation across all four levels. Question 5 will be addressed by examining the incremental variance explained by GDP-PPP after removing variability associated with household wealth and maternal education as well as the control variables. This question focuses on variation at level 4 attributable to GDP-PPP, net of household wealth and maternal education influences.

In preparation for the analysis, we subtracted the overall means of the following variables from their observed values: child age in months, maternal age in years, log of GDP-PPP and years of education. Rescaling provides more realistic modeled estimates for the average levels of child health without affecting the regression coefficients.

Results

Table 1 presents summary information on the total sample. There are 42 countries, 17,303 clusters,

Table 1
Sample characteristics

Level 4, countries[a] (*n*)	(42)
Number of clusters, *M* (min/max)	412.21 (99/1404)
GDP95-PPP, *M* (SD)	2146 (1744)
Level 3, clusters (*n*)	(17,303)
Number of households, *M* (min/max)	9.77 (1/179)
Level 2, households (*n*)	(150,318)
Number of children, *M* (min/max)	1.40 (1/12)
Household wealth, *M* (SD)	−0.12 (0.96)
Level 1, children (*n*)	(210,190)
Maternal education in years, *M* (SD)	4.23 (4.47)
Maternal age in years, *M* (SD)	28.01 (6.95)
% Male children	50.7
Child age months, *M* (SD)	25.51 (16.48)
Child weight for age (*z*-scores), *M* (SD)	−1.09 (1.37)
Child height for age (*z*-scores), *M* (SD)	−1.39 (1.57)

[a]Armenia, Bangladesh, Benin, Bolivia, Brazil, Burkino Faso, Cameroon, Central African Republic, Chad, Columbia, Comoros, Dominican Republic, Egypt, Ethiopia, Gabon, Ghana, Guatemala, Guinea, Haiti, India, Ivory Coast, Kazakhstan, Kenya, Kyrgyz Republic, Madagascar, Malawi, Mali, Mozambique, Namibia, Nepal, Nicaragua, Niger, Nigeria, Peru, Rwanda, Tanzania, Togo, Turkey, Uganda, Uzbekistan, Zambia, Zimbabwe.

Table 2
Multilevel null models of child weight and height for age

Characteristic	Weight for age coef. (SE)	Height for age coef. (SE)
Null model		
Intercept	−0.9817 (0.0771)	−1.2532 (0.0656)
Random effects variances		
Level 4 (country) σ^2_{v0}	0.2487 (0.0545)	0.1795 (0.0394)
Level 3 (clusters) σ^2_{u0}	0.1262 (0.0031)	0.1810 (0.0044)
Level 2 (household) σ^2_{w0}	0.2521 (0.0061)	0.2979 (0.0085)
Level 1 (child) σ^2_{e0}	1.2273 (0.0064)	1.7886 (0.0093)
Total	1.8543	2.4447
$-2\times$ loglikelyhood (IGLS)	687,092.3	760,160.9
Random effects variances expressed as a %		
Level 4, country	13.41	7.34
Level 3, clusters	6.81	7.40
Level 2, household	13.59	12.17
Level 1, child	66.19	73.09
Total	100.0	100.0

150,318 households and 210,190 children. The average level of GDP-PPP ($2146) is low. Within the 150,318 households, there are 164,084 women aged 15–49 years with children. The average weight and height for age of children are more than one standard deviation below US norms.

Table 2 shows the multilevel null or variance models to address question 1. Between-country variation associated with child weight and height for age is 13.41% and 7.34%, respectively. Within countries, about 20% of the variability of child health is associated with sampling clusters and households combined, with the latter accounting for about two-thirds of this variation. Most of the variation in child health is associated with between-child differences.

Influence of economic development, household wealth and maternal education

The final multilevel regressions appear in Table 3. The fixed effects coefficients for GDP-PPP, household wealth and maternal education with child health are all statistically significant, relatively large and comparable in size (question 2). The quadratic term for education is significant at $p<0.05$: it suggests that the association between child weight for age and maternal education gains strength at higher levels of education (0.00064). The strength of association between child height for age and maternal education squared is the same order of magnitude (0.00064) but falls short of statistical significance ($p=0.077$) because the estimate is less reliable.

In Table 3, all of the random effects regression coefficients at the country level (σ_{v2}^2, σ_{v3}^2, σ_{v4}^2 and σ_{v5}^2) are statistically significant (question 3). This indicates that both the strength and form of association between household wealth, maternal education and child health exhibits substantial cross-national variability. Fig. 1 graphs these relationships. The strong positive association between child health and household wealth is evident in Fig. 1(a) and (b). Also evident is cross-national variation in the form of this association. For example, the bolded upper curve in graph (a) represents the regression of child weight for age on wealth in Egypt: at higher levels of household wealth, the association is stronger. In contrast, the regression of child weight for age on wealth in Mali, lower curve in Fig. 1(a), shows the opposite: at higher levels of household wealth, the association is weaker, consistent with the idea of diminishing returns to health at upper levels of wealth. The positive association between child health and maternal education is evident in

Table 3
Multilevel regressions of child weight and height for age on all covariates[a]

Characteristic	Weight for age coef. (SE)	Height for age coef. (SE)
Fixed effects coefficients		
Intercept	−1.30437 (0.05215)***	−1.56169 (0.04570)***
Study variables		
Level 4 (country)		
Log GDP-PPP	0.39509 (0.07043)***	0.33966 (0.06138)***
Level 3 (cluster)	NA	NA
Level 2 (household)		
Household wealth	0.21308 (0.01115)***	0.24793 (0.01212)***
Household wealth squared	−0.00048 (0.00453)	0.00019 (0.00490)
Level 1 (child)		
Maternal education	0.03307 (0.00257)***	0.03661 (0.00321)***
Maternal education squared	0.00064 (0.00029)*	0.00064 (0.00036)
Control variables		
Level 1 (child)		
Maternal age in years	0.00502 (0.00045)***	0.00965 (0.00053)***
Maternal age in years squared	−0.00032 (0.00005)***	−0.00049 (0.00006)***
Male child	−0.05301 (0.00495)***	−0.08117 (0.00589)***
Child age in months	−0.02073 (0.00016)***	−0.02878 (0.00020)***
Child age in months squared	0.00121 (0.00001)***	0.00120 (0.00001)***
Random effects variances		
Level 4 (country)		
σ^2_{v0}: intercept	0.11103 (0.02461)***	0.08353 (0.01876)***
σ^2_{v2}: wealth	0.00391 (0.00111)***	0.00438 (0.00130)***
σ^2_{v3}: education	0.00020 (0.00006)***	0.00032 (0.00009)***
σ^2_{v4}: wealth^2	0.00058 (0.00018)**	0.00062 (0.00021)**
σ^2_{v5}: education^2	0.0000016 (0.0000007)*	0.0000028 (0.0000011)*
Level 3 (cluster) σ^2_{u0}	0.06616 (0.00213)***	0.09507 (0.00301)***
Level 2 (household) σ^2_{w0}	0.26348 (0.00528)***	0.33955 (0.00742)***
Level 1 (child) σ^2_{e0}	1.02942 (0.00544)***	1.47824 (0.00777)***
−2 × loglikelyhood (IGLS)	654,873.9	727,084.3

*$p<0.05$.
**$p<0.01$.
***$p<0.001$.

Figs. 1(c) and (d). On average, the association between child health and maternal education increases in strength at upper levels of education. This is well illustrated in Ghana—bolded lower curve in Fig. 1(c). Once again, this is not always the case. The regression of child weight for age on maternal education in the Dominican Republic—bolded upper curve in Fig. 1(c)—is flat.

Table 4 presents the amount of variation in child health explained by our multilevel models (question 4). These estimates are computed from the squared correlations between observed values of child weight and height for age and predicted values derived from nested multilevel models. The full models explain 19.036% and 17.131% of child weight and height for age, respectively. After removing the influence of the control variables, 12.234% and 8.940% of the variances are explained by the study variables. Among the study variables, GDP-PPP accounts for the largest amount of unique variability in child weight for age (4.251%) followed by maternal education (2.625%) and household wealth (2.226%). In child height for age, all three variables explain a similar amount of variation (~2.0%).

About one-quarter of the variation attributable to the study variables is shared (i.e., 3.132/12.234 and 2.633/8.940). Because the measure of household wealth is constructed separately within each country and independent of GDP-PPP, this shared variation

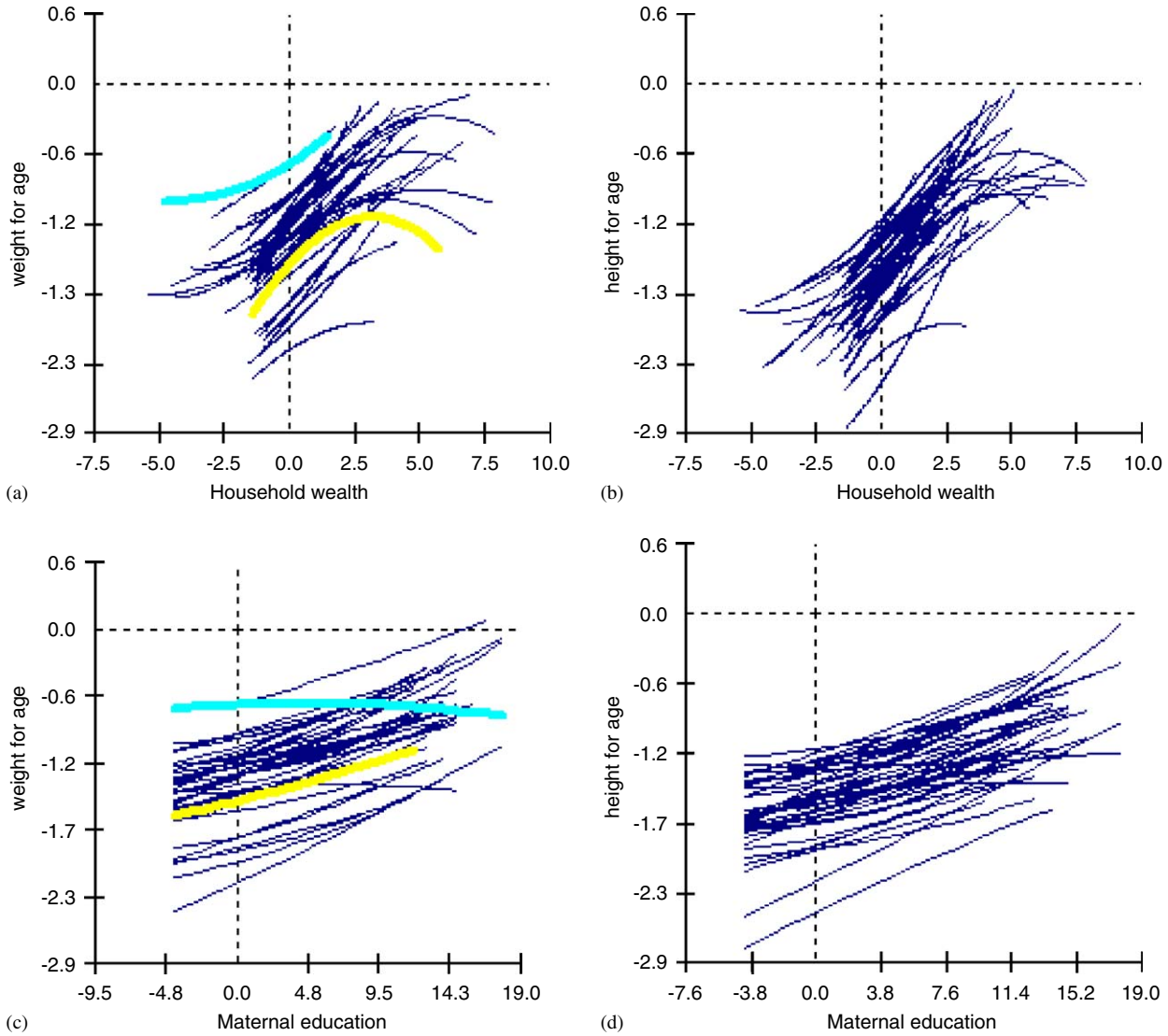

Fig. 1. Country-level regressions of weight (left side, a and c) and height (right side, b and d) for age on maternal education (a and b) and household wealth (c and d).

Table 4
Summary of explained variance

Variables	Weight for age	Height for age
Total explained variance	19.036	17.131
Control variables	6.802	8.191
Study variables	12.234	8.940
Study variables—unique (unadjusted)		
GDP-PPP	4.251 (5.879)	1.991 (3.024)
Household wealth	2.226 (4.108)	2.293 (4.094)
Maternal education	2.625 (5.323)	2.023 (4.454)
Study variables—shared	3.132	2.633

( ) Explained variance when the other two study variables are excluded.

arises from the separate overlap of maternal education with GDP-PPP and maternal education with household wealth. To illustrate how shared variance is distributed among the study variables, they were modeled one at a time to produce unadjusted estimates of explained variation shown in the brackets. The unadjusted estimates for maternal education explain almost as much or more variation in child health than GDP-PPP.

Because GDP-PPP is measured at the country level, all of the variability it explains in child health is associated with cross-national differences (question 5). Based on the null model estimates of cross-national variability in Table 2, the amount of

unique variation attributable to GDP-PPP is 31.7% (4.251/13.41) of weight for age and 27.1% (1.991/7.34) of height for age.

Discussion

In a sample of 42 developing countries, this study indicates that 13% and 7% of the variability in child health as measured by weight and height for age in standard deviation units is attributable to between-country differences. The cross-national difference in these two estimates likely reflects differences between the two indicators in their responsiveness to acute environmental influences early in the life course. For example, child weight for age will be susceptible very quickly to environmental influences such as acute infectious disease and poor nutritional intake (Sereebutra, Solomons, Aliyu, & Jolly, 2006). In contrast, child height for age is more likely to represent a long run measure of health or chronic undernourishment (Attanasio, Gomez, Rojas, & Vera-Hernandez, 2004). We suspect that between-country differences in acute disease levels combined with different latencies for influencing child weight and height, are responsible for these different variance estimates.

To our knowledge, this is the first study to estimate between-country variability in child growth using a multilevel approach. Pradham, Sahn, and Younger (2003) used aggregate data from both developed (OECD) and developing countries and estimated that cross-national differences accounted for about 32% of the variation in child height for age. The difference between our estimate and this one (i.e., 7% versus 32%) highlights the enormous gap in child health between the developing and developed world.

Although most of the overall variability of child health within countries is associated with between-child differences, about 20% of the total is associated with households and sampling clusters. The relatively large amount of variation associated with the sampling clusters (~7.0%) suggests that their influence might be explored further. This could be done by aggregating the characteristics of participants within clusters to create 'contextual' variables. The cautionary note is that these clusters were formed to increase survey efficiency; that nothing is known about their correspondence to meaningful 'neighborhoods'; and that they have been studied rarely (Montgomery & Hewett 2005).

Influence of economic development level, household wealth and maternal education

Economic development level, household wealth and maternal education have substantial and distinctive influences on child health. Level of economic development is an important factor underlying between-country differences in child health. This underscores its potential for improving levels of population health and agrees with ecological studies in developing countries looking at child mortality and malnutrition (Filmer & Pritchett, 1999a; Pritchett & Summers, 1996; Shen & Williamson, 2001; Smith & Haddad, 2001; Wang, 2003).

The strong association seen in this study between household wealth and child health replicates well established findings in both developed and developing countries (Keating & Hertzman, 1999; Leon & Walt, 2001). The interesting point here is that the distribution of household wealth within countries is as important to child height for age, as between-country differences in economic development. To our knowledge, this is the first study to make such a distinction. It is possible to do this because the wealth index is standardized separately to each country so that GDP-PPP and household wealth are measured orthogonally.

The unique influence of maternal education on child health is on a par with household wealth but falls short of GDP-PPP in explaining child weight for age. This focus on unique variation underestimates the potential impact of maternal education which can share influence with GDP-PPP as a between-country effect and with household wealth as a within-country effect. In fact, a substantial portion of maternal education's influence—about 50%—is shared with these two variables. The net influence of maternal education depends on presumed mechanisms of effect. It is arguable that women's education serves a positive feedback mechanism to both economic development, enhancing national productivity, and to household wealth, increasing resources available to families (Ranis et al., 2000). If this is true, then the influence on child health shared between GDP-PPP, household wealth and maternal education, is indeterminable. The empirical evidence distinguishing the effects of growth on education from the effects of education on growth is mixed: theoretical and methodological limitations of existing studies have yet to be addressed (Hannum & Buchmann, 2004).

Moderating effects of national contexts

This study also sought to quantify the moderating effects of national contexts on the relationship between household wealth, maternal education and child health. It is clear that countries exert moderating influences on the *strength* of these associations. Household wealth and maternal education are almost always positively associated with child health. However, there is substantial country-to-country variability in the strength of these associations and examples where the relationships between child health and maternal education are almost 'flat'.

National contexts also exert influence on the *form* of these associations. There is striking non-linearity in the relationship between household wealth and child health. In a number of countries, the decelerating strength of association between these two variables is consistent with the idea of diminishing returns to child health among wealthier households (Subramanian et al., 2002). In several other countries, the opposite occurs—increasing returns to child health among wealthier households. It is noteworthy that the fixed effect estimates for wealth squared (quadratic effects) are statistically non-significant. Allowing the quadratic effect for household wealth to vary randomly from country to country made it possible to identify these cross-national differences in the form of association between child health and household wealth.

Furthermore, there is an important difference in the pattern of associations between household wealth, maternal education and child health. Whereas the fixed effects estimates for household wealth squared are relatively small, non-significant and mixed in their direction of influence, the comparable fixed effects estimates for maternal education squared are relatively large, statistically significant (or approaching significance) and positive in direction. This suggests that the returns to child health may be consistently stronger at upper levels of maternal education. This pattern of association with child health is substantially different from those observed for economic development and household wealth where diminishing returns to health at upper levels are noted consistently. In the present study, these diminishing returns to child health at upper levels of wealth are reflected: (1) in the need to model child health on the log of GDP-PPP and (2) in the decelerating influence of household wealth on child health among more affluent households in many countries.

In sum, the results of this study run counter to reports questioning the impact of maternal education on child health (e.g., Basu, 1994; Desai & Alva, 1998; Hobcraft, 1993). On a comparative basis, the unique influence of maternal education on child health is arguably stronger than the unique influence of within-country distributions of household wealth. Second, it would be mistake to discount the influence of maternal education that is shared with GDP-PPP and household wealth. Researchers who do this are making causal assumptions about the timing and direction of effects attributable to these factors. Third, returns to child health appear to accelerate at higher levels of education—a pattern substantially different from that observed for household wealth.

DHS and multilevel modeling

In the present study, we have used multilevel regression to decompose the unique and shared variation in child health attributable to three interrelated influences—GDP-PPP, household wealth and maternal education—and to examine the impact of national contexts on these relationships. Answering these types of questions successfully require data sets such as the DHS which integrate macro (country-level) and micro (individual-level) information. Such data sets are invaluable; they open up the study of contextual effects, making it possible to distinguish between population-level effects (factors that influence average levels of health in the population) and individual-level effects (factors that influence the distribution of health effects within populations). Further to this, these models can be used to examine and to understand the moderating influence of contexts (e.g., countries) on individual-level processes occurring within populations.

The findings from this study illustrate some of the advantages of both the DHS data set and the modeling approach. One, among developing countries, there is significant cross-national variation in average levels of child health. Although this has been well established for decades, the statistical approach quantifies the potential influence on child health associated with between-country differences in relation to other sources of influence such as households.

Two, maternal education has an impact on both the distribution of health within countries (i.e., within-country gradients for child health repre-

sented by their regression coefficients) as well as average levels of child health (i.e., between-country differences in the intercept). The impact of maternal education on average levels of child health in this study is inferred from the shared variance between GDP-PPP and maternal education. GDP-PPP is measured at the country level and its influence is restricted to country-level effects. Maternal education is measured at the individual level and its influence is not restricted; it can exert effects on persons and on populations. It is relatively straightforward to extend the partitioning of unique and shared variance for variables such as maternal education from general estimates such as those presented in this study, to specific estimates associated with particular levels (e.g., see Boyle & Lipman, 2002). This analytical capability is extremely useful for understanding the contribution of variables measured at the individual level, to population-level effects.

Three, as noted in the methods section, the random effects co-variances—associations between random effects intercepts at the country level and random effects regression coefficients at the country level—were small in magnitude, statistically non-significant and excluded from the model. Although little was made of this 'finding'—it was the by-product of identifying a suitable model—it serves as another example of the potential usefulness of the DHS data set and multilevel modeling. Negative associations between country-level intercepts and slopes would provide evidence that countries with wider health disparities based on wealth and education have lower levels of health. Although this was not the case in the present study, the opportunity to test such a hypothesis is only available in data such as the DHS using multilevel approaches.

Four, the strength and form of associations between child health and household wealth and between child health and maternal education are subject to national influences. These national influences could be attributable to many different factors: governance, infrastructure, public health, culture, religion and so on. Although the present study cannot account for these moderating influences, other information contained in the DHS and World Development Indicator data base provide the basis for generating and testing specific hypotheses about these effects. Extending this idea of country differences, there are several examples where national context seems to exert a dramatic impact on within-country relationships between child health and household wealth. Multilevel modeling makes it possible to identify these countries and to test empirically the statistical meaningfulness of their differences with other countries. These extreme examples may represent rich sources of information for case studies.

This study has a number of limitations. One, the study is cross-sectional in design, and we are unable to tease apart the mechanisms of effect associated with our key variables. An understanding of causal processes is essential for generating useful policies to improve population health. Two, the countries included in this study represent a sample of convenience. Three, the amount of variation explained by our statistical models is relatively small. A contributing factor to this is the limited number of variables selected for study. Four, we have not examined the impact of income inequality on child health. This is a very important question to address but requires reliable and valid methods to measure income disparities in developing countries. Deaton (2003) has argued persuasively that poor conceptualization and measurement of income inequalities are behind the disparate findings reported in the literature. Five, we are unable to take into account genetic variation in child height and weight: this would surely explain substantial variability in our dependent variables and might account for some of our cross-national differences.

Choosing optimal strategies for improving population health overall and equitable distributions of health within countries poses a serious challenge for governments with limited resources (Leon, Walt, & Gilson, 2001). Each of the strategies examined in our study—economic development level, household wealth and maternal education—offer considerable leverage for improving child health at both population and individual levels. Considering these strategies in isolation from one another is a mistake: there is substantial overlap (shared variance) in the effects they produce on child health and these effects are subject to modification by the national contexts in which they operate. We hope that our approach to analysing the influence of these strategies provides new insights into the opportunities for improving child health in the developing world.

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References

Alderman, H., Appleton, S., Haddad, L., Song, L., & Yisehac, Y. (2001). Reducing child malnutrition: How far does income growth take us? CREDIT Research Paper (No. 01/05). Centre for Research in Economic Development and International Trade, University of Nottingham.

Anand, S., & Ravallion, M. (1993). Human development in poor countries: On the role of private incomes and public services. *Journal of Economic Perspectives*, *7*, 133–150.

Attanasio, O., Gomez, L. C., Rojas, A. G., & Vera-Hernandez, M. (2004). Child health in rural Columbia: Determinants and policy interventions. *Economics and Human Biology*, *2*, 411–438.

Barrera, R. (1990). The role of maternal schooling and its interaction with public health programs in child health production. *Journal of Developmental Economics*, *32*, 69–91.

Basu, A. M. (1994). Maternal education, fertility and child mortality: Disentangling verbal relationships. *Health Transition Review*, *4*, 207–215.

Bollen, K. A., Glanville, J. L., & Stecklov, G. (2002). Economic status proxies in studies of fertility in developing countries: Does the measure matter? *Population Studies*, *56*, 81–96.

Boyle, M. H., & Lipman, E. L. (2002). Do places matter? Socioeconomic disadvantage and child problem behaviour in Canada. *Journal of Consulting and Clinical Psychology*, *70*, 378–389.

Buchmann, C. (1996). The debt crisis, structural adjustment and women's education: Implications for status and social development. *International Journal of Comparative Sociology*, *37*, 5–30.

Caldwell, J. C. (1979). Education as a factor in mortality decline: An examination of Nigerian data. *Population Studies*, *33*, 395–413.

Caldwell, J. C. (1994). How is greater maternal education translated into lower child mortality? *Health Transition Review*, *4*, 224–229.

Deaton, A. (2003). Health, inequality, and economic development. *Journal of Economic Literature*, *XLI*, 113–158.

Desai, S., & Alva, S. (1998). Maternal education and child health: Is there a strong causal relationship? *Demography*, *35*, 71–81.

Dreze, J., & Sen, A. (1989). *Hunger and public action*. Oxford: Clarendon Press.

Filmer, D., & Pritchett, L. H. (1999a). The impact of public spending on health. Does money matter. *Social Science and Medicine*, *49*, 1309–1323.

Filmer, D., & Pritchett, L. (1999b). The effect of household wealth on educational attainment: Evidence from 35 countries. *Population Development Review*, *25*, 85–120.

Filmer, D., & Pritchett, L. H. (2001). Estimating wealth effects without expenditure data—or tears: An application to educational enrollments in states of India. *Demography*, *38*, 115–132.

Gwatkin, D. R. (2001). Poverty and inequality in health within developing countries: Filling the information gap. In D. A. Leon, & G. Walt (Eds.), *Poverty, inequality and health: An international perspective* (pp. 217–246). Oxford: Oxford University Press.

Hadden, K., & London, B. (1996). Educating girls in the Third World: The economic benefits. *International Journal of Comparative Sociology*, *37*, 31–46.

Hannum, E., & Buchmann, C. (2004). Global educational expansion and socio-economic development: An assessment of findings from the social sciences. *World Development*, *33*, 333–354.

Hobcraft, J. N. (1993). Women's education, child welfare and child survival: A review of the evidence. *Health Transition Review*, *3*, 159–175.

Houweling, T. J., Kunst, A. E., & Mackenbach, J. P. (2003). Measuring health inequality among children in developing countries: Does the choice of the indicator of economic status matter? *International Journal for Equity in Health*, *2*, 8–17.

Keating, D. P., & Hertzman, C. (Eds.). (1999). *Developmental health and the wealth of nations*. London: The Guilford Press.

Kreft, I., & deLeeuw, J. (1998). *Introducing multilevel modeling*. London: Sage Publications.

Leon, D. A., & Walt, G. (Eds.). (2001). *Poverty, inequality and health: An international perspective*. Oxford: Oxford University Press.

Leon, D. A., Walt, G., & Gilson, L. (2001). International perspectives on health inequalities and policy. *BMJ*, *322*, 591–594.

Macro International Incorporated. (1996). *Sampling Manual*. Calverton, Maryland: DHS-III Basic Documentation No. 6.

Montgomery, M. R., & Hewett, P. C. (2005). Urban poverty and health in developing countries: Household and neighborhood effects. *Demography*, *42*, 397–425.

Pradham, M., Sahn, D. E., & Younger, S. D. (2003). Decomposing world health inequality. *Journal of Health Economics*, *22*, 271–293.

Pritchett, L., & Summers, L. H. (1996). Wealthier is healthier. *Journal of Human Resources*, *31*, 841–868.

Ranis, G., Stewart, F., & Ramirez, A. (2000). Economic growth and human development. *World Development*, *28*, 197–219.

Rasbash, J., Steele, F., Browne, W., & Prosser, B. (2004). *A user's guide to MLwiN*. London: Institute of Education.

Schultz, T. P. (2002). Why governments should invest more to educate girls. *World Development*, *13*, 827–846.

Sen, A. (2001). Economic progress and health. In D. A. Leon, & G. Walt (Eds.), *Poverty, inequality and health: An international perspective* (pp. 333–345). Oxford: Oxford University Press.

Sereebutra, P., Solomons, N., Aliyu, M. H., & Jolly, P. E. (2006). Sociodemographic and environmental predictors of childhood stunting in rural Guatemala. *Nutrition Research*, *26*, 65–70.

Shen, C., & Williamson, J. B. (2001). Accounting for cross-national differences in infant mortality decline (1965–1991) among less developed countries: effects of women's status, economic dependency, and state strength. *Social Indicators Research*, *53*, 257–288.

Smith, L. C., & Haddad, L. (2001). How important is improving food availability for reducing child malnutrition in developing countries? *Agricultural Economics*, *26*, 191–204.

Snijders, T., & Bosker, R. (1999). *An introduction to basic and advanced multilevel modeling*. London: Sage Publications.

Subramanian, S. V., Belli, B., & Kawachi, I. (2002). The macroeconomic determinants of health. *Annual Review of Public Health*, *23*, 287–302.

Wagstaff, A. (2000). Socioeconomic inequalities in child mortality: Comparisons across nine developing countries. *Bulletin of the World Health Organization*, *78*, 19–29.

Wang, L. (2003). Determinants of child mortality in LDC's: Empirical findings from demographic and health surveys. *Health Policy*, *65*, 277–299.

World Bank. (2004). *World Development Indicators*. https://publications.worldbank.org/WDI/

World Health Organization. (1999). *World health report: Making a difference*. Geneva: WHO.